

FAUZIYYA A. AHMED

Data Science | Full-Stack Development | IoT & Robotics

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Portfolio Website: <https://fauziea.github.io/My-portfolio/>

PROFESSIONAL SUMMARY

Computer Science student specializing in **Data Science** with diverse practical expertise in **Full-Stack Development** and **IoT**. Experienced in engineering end-to-end solutions, from predictive ML models to autonomous robotics and scalable web platforms. Winner of the **Hult Prize** (On-Campus), demonstrating a proven ability to deliver technical innovation that solves real-world problems.

TECHNICAL SKILLS

Languages: Python (PCEP Certified), JavaScript (ES6+), PHP, C, SQL.

Web Development: MERN Stack (React.js, Node.js, Express.js), Next.js, Django, HTML5/CSS3, Tailwind CSS, Bootstrap.

Data Science & AI: TensorFlow/Keras, Scikit-learn, Pandas, NumPy, NLP (TF-IDF), Computer Vision (CNN/RNN), SHAP, Recommender Systems.

IoT & Cloud: Smart Systems Integration, CCTV/Video Analytics, REST APIs, Automation Workflows, AWS & IBM Cloud Services.

Platforms & CMS: WordPress, Moodle LMS, Shopify, Git/GitHub, GitHub Actions.

Tools & Databases: Power BI, MySQL, SQLite, VS Code, Postman, Streamlit, Figma.

Soft Skills: Technical Leadership, Public Speaking, Workshop Facilitation, Agile Collaboration, Strategic Planning.

EDUCATION

BSc (Hons) Computer Science | Albukhary International University, Malaysia

Jan 2023 – Present | **CGPA: 3.77/4.00**

- Honors:** Dean's List (Awarded every semester).
- Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Operating Systems, IoT, Machine Learning, Deep Learning, NLP.

EXPERIENCE

Web Developer Intern | Avidity International

Mar 2025 – Aug 2025

- Developed and managed Moodle LMS platforms, implementing structured learning paths and secure content access for diverse user cohorts.
- Optimized WordPress UI/UX, improving site responsiveness and implementing component animations to increase user engagement.
- Collaborated with cross-functional teams to troubleshoot backend issues and ensure 99.9% platform uptime.

Co-Founder & Lead Instructor | TechNexus (Student-Led Tech Initiative)

Jan 2025 – Present

- Leadership:** Directing a global initiative to democratize tech education; organized the "Swiss Innovation Challenge" team, presenting the "Poultrix" project.
- Instruction:** Designed and taught technical bootcamps (MERN Stack, Database Management, Intro to HTML) to 200+ students.
- Event Management:** Hosted a tech booth during the **Prime Minister of Malaysia's** official visit to AIU, showcasing student innovations to high-level dignitaries.

Library Associate | Albukhary International University Library

Aug 2024 – May 2025

- Manage library database systems and assist students with research resource allocation.
- Organize academic engagement events, including the "AIU Peers" study sessions for Probability & Statistics.

COO & Deputy Key Person | 3Zero Club & Po-Stitch (Social Business)

Jan 2024 – Present

- Co-manage "Po-Stitch," a sustainable social business focused on upcycling waste into fashion products.
- Ambassadorship:** Represented the initiative at the "Social Business Presentation" and managed a showcase booth during high-profile university events.
- Collaborated with German exchange students (Aug 2024) to expand the project's international reach.

PROJECTS

Deep Learning Handwriting Recognition System | *Python, TensorFlow/Keras, OpenCV*

Dec 2025

- Developed a CNN-based writer identification system achieving 87% test accuracy across 70 distinct handwriting styles.
- Built a custom preprocessing pipeline using OpenCV for line/word segmentation and patch-based data augmentation.
- Deployed the trained model as a REST API using FastAPI, enabling real-time inference on uploaded document images.

IoT Line-Following & Remote Monitoring Robot | *Python, C++, ESP32, MQTT*

Dec 2025

- Engineered an autonomous robot capable of precise line following using infrared sensor arrays.
- Integrated a remote monitoring dashboard allowing real-time telemetry tracking and status updates via Wi-Fi/Bluetooth.
- Designed the control logic to handle edge cases and obstacle avoidance for robust navigation.

Unified Smart Lighting & CCTV Integration Platform (Final Year Project) | *Python, REST APIs, Cloud Integration*

Oct 2025 – Present

- Building a brand-agnostic IoT platform unifying lighting systems and CCTV for automation and energy optimization.
- Designed a four-layer system architecture (interface, processing, data, device) with secure communication and scalable APIs.

Personal Library & Book Recommendation System | *Python, Streamlit, NLP (TF-IDF)*

Nov 2025

- Built a full reading-tracking and hybrid recommendation system (metadata + TF-IDF) using Streamlit.
- Integrated external datasets for personalized genre and preference matching.

TechNexus Official Website | *React.js, Tailwind CSS, GitHub Pages*

Sep 2025

- Deployed a responsive web platform to showcase initiative programs, missions, and student services to a global audience.
- Implemented a modern, component-based UI ensuring fast load times and mobile compatibility.

Customer Churn Prediction | *Python, Scikit-learn, SHAP, SMOTE*

Jun 2025 – Jul 2025

- Built a churn prediction model using classification algorithms with engineered features for improved performance.
- Used SHAP and confusion matrices to interpret results and deliver actionable business insights.

Traffic Congestion Prediction (Best Paper Award) | *Python, Scikit-learn, Research*

Mar 2025 – Jun 2025

- **Award:** Won **Best Paper Award** for "Predictive AI Models for Traffic Prediction" (June 30, 2025).
 - Investigated classical ML models vs. complex neural networks for short-term traffic forecasting.
 - Engineered temporal features to achieve high-accuracy predictions, validated through rigorous cross-validation.
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AWARDS & COMPETITIONS

- **Winner, On-Campus Hult Prize (2025):** Secured 1st place for social entrepreneurship; led team in ideating scalable business solutions for global impact.
 - **Best Project Award (Social Cluster) (Feb 2024):** Recognized for the most impactful project in Sustainability and Society.
 - **Grand Heritage Ambassador (Feb 2025):** Awarded during AIU Cultural Week for outstanding representation and community leadership.
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LEADERSHIP & VOLUNTEERING

- **Moderator, Kedah Maker Fair:** Hosted sessions and facilitated discussions at a state-level technology fair.
 - **Moderator, AIU Center of Reference:** Hosted the "From Dawn to Dusk" alumni talk series.
 - **Train-the-Trainer, Young Innovators Challenge:** Certified to mentor high school students in innovation and design thinking (April 2025).
 - **Participant, Google I/O Extended & Build with AI:** Attended developer conferences in Georgetown/Penang to stay updated on Generative AI and Cloud trends.
 - **Volunteer, Malaysia Volunteerism Program:** Engaged in community development at Kampung Pantai Jemai (June 2024).
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CERTIFICATIONS

- PCEP: Certified Entry-Level Python Programmer
- IBM Data Fundamentals
- IBM Cloud Essentials
- Kaggle: Advanced SQL, Data Visualization, Intro to Machine Learning
- Qubit x Microsoft Azure: Quantum Winter School
- C for Beginners (Great Learning Academy)